QQ-P-35B
April 5, 1973
SUPERSEDING
Fed. Spec. QQ-P-35A
June 20, 1968

# FEDERAL SPECIFICATION

# PASSIVATION TREATMENTS FOR CORROSION-RESISTING STEEL

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration for use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers six types of passivation treatments for austenitic, ferritic, martensitic and precipitation hardening corrosion-resisting steels.

1.2 Classification.

1.2.1 Types. Passivation treatments shall be of the following types:

Type I - Low temperature.
Type II - Medium temperature.
Type III - High temperature.
Type IV - For steels containing large amounts (0.15 percent) of sulfur or selenium.
Type V - Anodic - For high carbon martensitic (440) steels.
Type VI - Low temperature (optional).

1.2.2 For parts made of austenitic 300 series corrosion-resisting steels which do not contain relatively large amounts of sulfur and selenium, Types I, II, III or VI shall be used (see 6.2).



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1.2.3 For parts made of ferritic 400 series and martensitic 400 series corrosion-resisting steels which do not contain relatively large amounts of sulfur and selenium and for austenitic (for example ASTM A286) and martensitic (for example 17-4PH) precipitation hardening steels, Types I, II or III shall be used (see 6.2).

1.2.4 For parts made of corrosion-resisting steels containing relatively large amounts (0.15 percent) of sulfur or selenium (for example 303, 303Se, 347Se, 416, 416Se, 430F and 430FSe), Type IV shall be used (see 6.2).

1.2.5 For parts made of high carbon martensitic (440) corrosion-resisting steels, Type V shall be used (see 6.2). For 440F and 440Se, Type IV may be used.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

O-N-350 - Nitric Acid, Technical.
O-S-595 - Sodium Dichromate, Dihydrate, Technical.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Speicifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-S-5002 - Surface Treatments and Inorganic Coatings for Metal Surfaces of Weapons Systems.



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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-753 - Corrosion-Resistant Steel Parts: Sampling, Inspection and Testing for Surface Passivation.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

B117 - Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103.)

3. REQUIREMENTS

3.1 Material. The chemicals and reagents used for passivation of corrosion-resisting steel parts shall produce passivated surfaces which meet the requirements of this specification.

3.2 Pretreatment. The pretreatment methods and procedures used prior to the passivation treatment shall be in accordance with MIL-S-5002, shall leave the surface of the parts free of oil, grease, rust, scale and other foreign matter and shall have no deleterious effect on material properties. Particular attention shall be given to the pretreatment cleaning of 400 series steels to be certain that all surface oxidation, including rust and heat treat scale, is completely removed.

3.3 Passivation treatment. Corrosion-resisting steel parts shall be passivated by immersing them in one of the following aqueous solutions (see 6.2) and maintaining them at the temperature range and for the time shown in Table I. The parts shall be completely immersed in the solution to avoid severe etching which would otherwise occur above the liquid line.

3.3.1 Types I, II and III. Types I, II and III solutions shall contain 20 to 25 percent by volume of nitric acid ($HNO_3$ - sp. gr. 1.42) in accordance with O-N-350 and 2 to 3 percent by weight of sodium dichromate ($Na_2Cr_2O_7 \cdot 2H_2O$) in accordance with O-S-595.

3.3.2 Type IV. Type IV solution shall contain 38 to 42 percent by volume of nitric acid in accordance with O-N-350 and 2 to 3 percent by weight of sodium dichromate in accordance with O-S-595.



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3.3.3. Type V. Type V solution shall contain 20 to 25 percent by volume of nitric acid in accordance with O-N-350 and 2 to 3 percent by weight of sodium dichromate in accordance with O-S-595. The steel parts shall be made the anode in the solution operated at a potential of 3 to 5 volts.

3.3.4 Type VI. Type VI solution shall contain 20 to 50 percent by volume of nitric acid in accordance with O-N-350. No sodium dichromate shall be used.

TABLE I. Passivation treatments

| Type | Temperature (degrees F.) | Time, minimum (minutes) 1/ |
|---|---|---|
| I | 70 - 90 | 30 |
| II | 120 - 130 | 20 |
| III | 145 - 155 | 10 |
| IV | 70 - 120 | 30 |
| V | 120 - 130 | 2 |
| VI | 70 - 90 | 30 |

1/ Caution: Care must be exercised as excessive time may damage the part.

3.4 Water rinse. Immediately after removal from the passivating solution the parts shall be thoroughly rinsed in clean water (70°F. - 200°F.).

3.5 Chromate treatment. Within one hour after the water rinse (3.4), all ferritic and martensitic parts shall be immersed in an aqueous solution containing 4 to 6 percent by weight of sodium dichromate in accordance with O-S-595, at a temperature of 140°F. to 160°F. for a period of 30 minutes. This immersion shall be followed by a rinse in clean water. The parts shall then be thoroughly dried.

3.6 Finish. The passivated parts shall exhibit a clean surface and shall show no etching, pitting or frosting when examined. A slight discoloration will be allowed.

3.7 Staining. The passivated parts shall not exhibit staining attributable to the presence of free iron particles imbedded in the surface when subjected to the water immersion test or the high humidity test and the salt spray test. When the copper sulfate test is used, passivated parts shall not exhibit copper deposits.

3.8 Workmanship. The passivated parts shall be free of iron contamination and other foreign materials which could adversely affect the suitability, use or life of the passivated part.



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3.9 Solution analysis. The solutions used in these passivation treatments shall be periodically analyzed to verify that the amounts of nitric acid and sodium dichromate are within the specified requirements.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the Supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection provisions.

4.2.1 Lot. A lot shall consist of passivated parts of similar alloy or manufacturing methods that are pretreated and passivated in one day's production. When the quantity of passivated parts in one day's production does not warrant daily testing, the lot size shall be as agreed upon by the procuring activity and the contractor (see 6.2).

4.2.2 Sampling.

4.2.2.1 Sampling for examination. Samples shall be taken at random from each lot in accordance with MIL-STD-105, Inspection Level S-4. The Acceptable Quality Level (AQL) shall be 1.0 percent defective.

4.2.2.2 Sampling for tests. Samples shall be taken at random from each lot in accordance with MIL-STD-105, Inspection Level S-3. The AQL shall be 1.0 percent defective. Identically processed panels may be used for test purposes in lieu of large parts. The test panels shall be randomly selected from the same material used to fabricate the parts and shall be randomly distributed throughout the lot during processing.

4.3 Examination. Samples selected in accordance with 4.2.2.1 shall be visually examined for etching, pitting or frosting (3.6) and workmanship (3.8).

4.4 Test methods.

4.4.1 Water immersion and high humidity tests. Samples selected in accordance with 4.2.2.2 shall be subjected to either a water immersion test or a high humidity test to determine conformance with 3.7.

4.4.1.1 Water immersion test. The water immersion test shall be conducted in accordance with Method 100 of MIL-STD-753.



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4.4.1.2 High humidity test. The high humidity test shall be conducted as follows:

a. Parts shall be cleaned by immersing them in acetone or methyl alcohol, then swabbing them with clean gauze saturated with acetone or methyl alcohol and drying in an inert atmosphere or dessicated container.

b. The cleaned and dried parts shall be subjected to 95-100 percent humidity at 100°F. in a suitable humidity cabinet for 24 hours.

4.4.2 Salt spray test. Samples selected in accordance with 4.2.2.2 shall be subjected to a salt spray test conducted in accordance with ASTM B117 for a minimum of two hours using a 5 percent salt spray solution, to determine conformance with 3.7.

4.4.3 Copper sulfate test. This test applies to 300 series, corrosion-resisting steels containing relatively large amounts of sulfur or selenium (303, 303Se and 347Se) and should be used in lieu of the salt spray test for these steels. Samples selected in accordance with 4.2.2.2 shall be subjected to a copper sulfate test conducted in accordance with Method 102 of MIL-STD-753, to determine conformance with 3.7.

4.5 When specified (see 6.2), a certificate of the quantitative analysis (3.9) of the passivating solution shall be furnished the contracting officer.

4.6 Rejection and retest. Any lot failing to meet the requirements specified herein shall be rejected. A rejected lot may be resubmitted for inspection provided that the defective parts have been re-pretreated per 3.2, as necessary, repassivated and re-tested. Re-passivated lots shall be subjected to tightened inspection in accordance with MIL-STD-105.

## 5. PREPARATION FOR DELIVERY

(This section is not applicable to this specification.)

## 6. NOTES

6.1 Intended use. The passivation treatments provided by this specification are intended to improve the corrosion resistence of parts made from austenitic, ferritic and martensitic corrosion-resisting steels of the 300 and 400 series and precipitation hardening corrosion-resisting steels.

6.1.1 During the processing operations such as forming, machining, tumbling and lapping, iron particles or other foreign particles may become smeared over or imbedded into the surface of corrosion-resisting steel parts. These particles must be removed or they will appear as rust or stain spots. This condition may be prevented by immersing the parts in a solution of nitric acid and sodium dichromate, which will dissolve the particles and leave a thin transparent film over the restored surface.



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6.1.2 This specification is not intended for black oxide coating of parts to be used for photographic and optical instruments; for such applications refer to MIL-C-13924.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Nomenclature and material of parts.
(c) Type of passivation treatment (3.3).
(d) Lot size, if other than one day's production (4.2.1).
(e) When quantitative analysis certificate is required (4.5).

MILITARY CUSTODIANS:
Army - WC
Navy - SH
Air Force - 11

Preparing Activity:
Army - WC

Project No. 53GP-0052

Reviewer Activities:
Army - AV, EL, GL, MI, MR, MU
Navy - AS, OS
Air Force - 80, 82
DSA - IS

Civil Agency Coordinating Activity:
GSA-FSS

User Activities:
Army - AT, ME
Navy - None
Air Force - None



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

QQ-P-35B - Passivation Treatments for Corrosion-Resisting Steel

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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